```
Loading required package: pacman

Rows: 440877 Columns: 110
── Column specification ────────────────────────────────────────────────────────
Delimiter: ","
chr (16): contractid, state, county, org_type, plan_type, partd, snp, eghp, ...
dbl (94): planid, fips, n_nonmiss, avg_enrollment, sd_enrollment, min_enroll...

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Question 1

A tibble: 6 × 5

year	Mean Star Rating	Mean Enrollment	Mean Market Share	Plan Count
<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
2010	2.970	345.970	0.012	60519
2011	3.255	388.308	0.015	55249
2012	3.360	410.649	0.017	58169
2013	3.549	406.733	0.016	64174
2014	3.806	451.114	0.018	58636
2015	3.962	482.571	0.019	60819

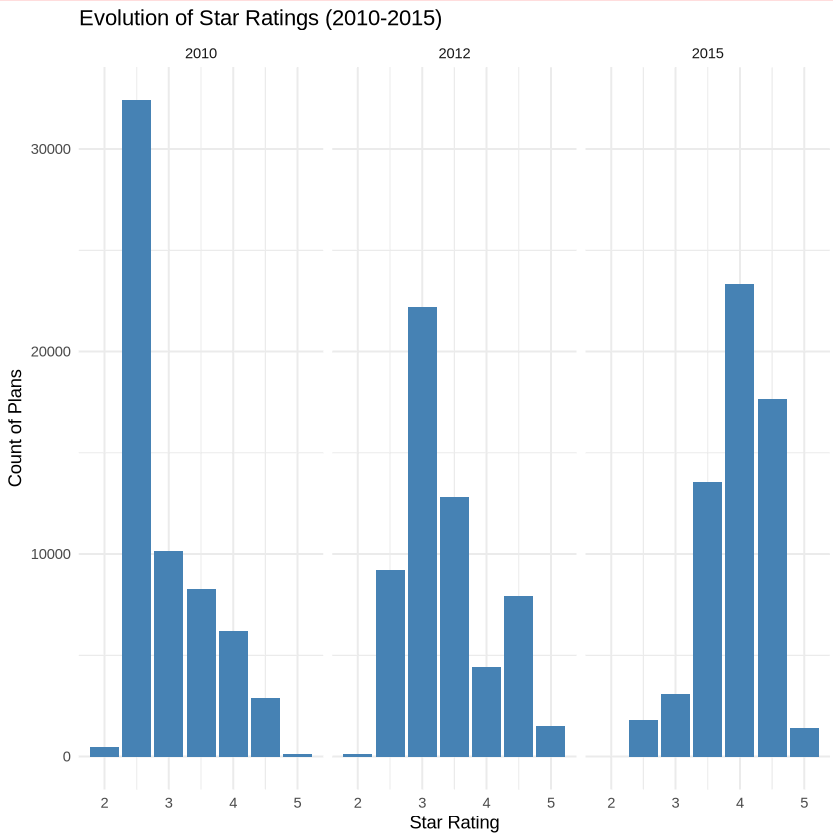
Question 2

A tibble: 6 × 4

year	Mean Enrollment	Mean Market Share	Count of Plans
<dbl>	<dbl>	<dbl>	<dbl>
2010	107.258	0.009	49431
2011	167.899	0.013	12754
2012	172.734	0.012	9077
2013	213.504	0.017	3661
2014	257.596	0.016	3708
2015	314.210	0.009	4680

Question 3

```
Warning message:
“Removed 63188 rows containing non-finite values (`stat_count()`).”
```



Between 2010 and 2015, the distribution of MA star ratings saw a large rightward shift (more plans with higher star ratings). In 2010, the market was dominated by lower-rated plans, with 2.5 stars being the most frequent at 30,000+ occurrences. By 2012, the distribution began to center with 3.0 being the most common rating. 2015 shows how the distrubution then shifted to a high-quality skew where 4.0 plans were the highest count of plans, while the count of 2.5 star plans dropped to below 5,000.

Question 4

```
Regression Results: Market Share vs. Star Rating (2010–2015)
=====================================================================
                              Dependent variable:
             -------------------------------------------------------
                                   Market Share
               2010      2011      2012      2013     2014     2015
               (1)       (2)       (3)       (4)      (5)      (6)
---------------------------------------------------------------------
3.0 Stars    0.013***  0.035***  0.031***  0.062*** 0.027*** 0.027***
             (0.002)   (0.002)   (0.003)   (0.003)  (0.006)  (0.005)

3.5 Stars     0.001    0.019***  0.013***  0.038*** 0.027*** 0.039***
             (0.002)   (0.003)   (0.003)   (0.003)  (0.006)  (0.004)

4.0 Stars    -0.003     0.003   -0.025*** 0.017***  0.007   0.011***
             (0.002)   (0.003)   (0.004)   (0.003)  (0.006)  (0.004)

4.5+ Stars   -0.007*** -0.009*** -0.013*** 0.022***  0.007   0.023***
             (0.003)   (0.003)   (0.003)   (0.003)  (0.006)  (0.004)

Constant     0.063***  0.070***  0.079***  0.049*** 0.064*** 0.061***
             (0.001)   (0.002)   (0.003)   (0.003)  (0.006)  (0.004)

---------------------------------------------------------------------
Observations  19,072    19,260    20,895    23,755   22,793   22,949
R2             0.004     0.019     0.023     0.021    0.007    0.008
Adjusted R2    0.004     0.019     0.023     0.021    0.006    0.008
=====================================================================
Note:                                     *p<0.1; **p<0.05; ***p<0.01
```

Question 5

A tibble: 5 × 2

Star Rating	Number of Plans Rounded Up
<chr>	<int>
3.0 Stars	3808
3.5 Stars	4524
4.0 Stars	4077
4.5 Stars	773
5.0 Stars	30

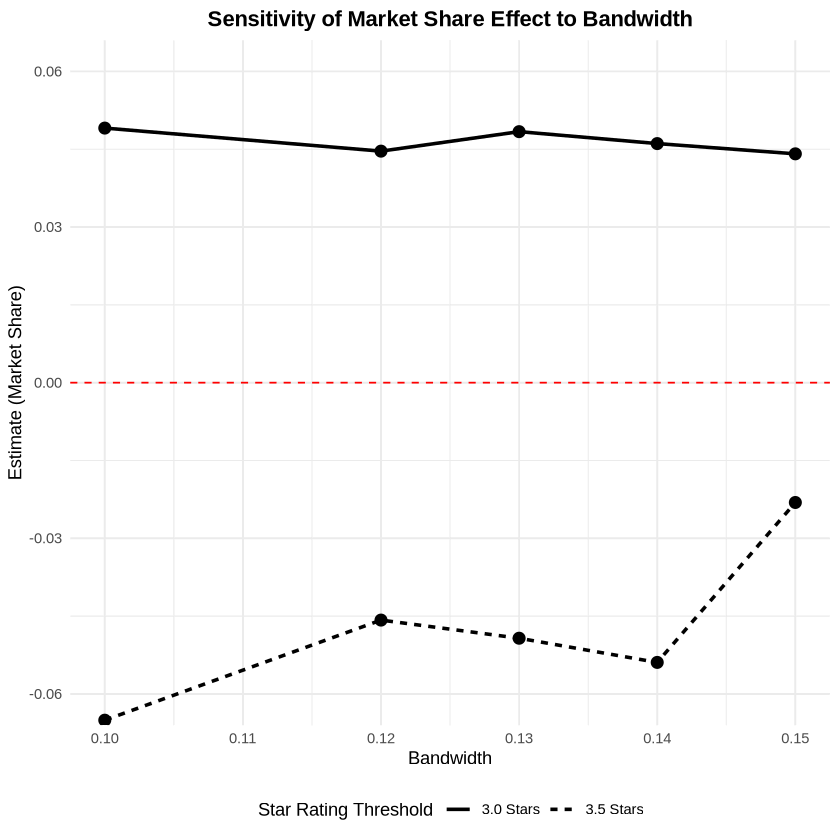
Question 6

Table 6: RD Estimates of Star Rating on Market Share

	Dependent variable:	
	Market Share	
	3.0 Stars	3.5 Stars
	(1)	(2)
Rounded	0.046***	-0.049***
	(0.007)	(0.009)
Running Score	-0.589***	0.522***
	(0.039)	(0.070)
Rounded x Score	0.656***	-0.552***
	(0.093)	(0.097)
Constant	0.022***	0.110***
	(0.003)	(0.007)
Observations	5,984	2,172
R2	0.038	0.026
Adjusted R2	0.037	0.024

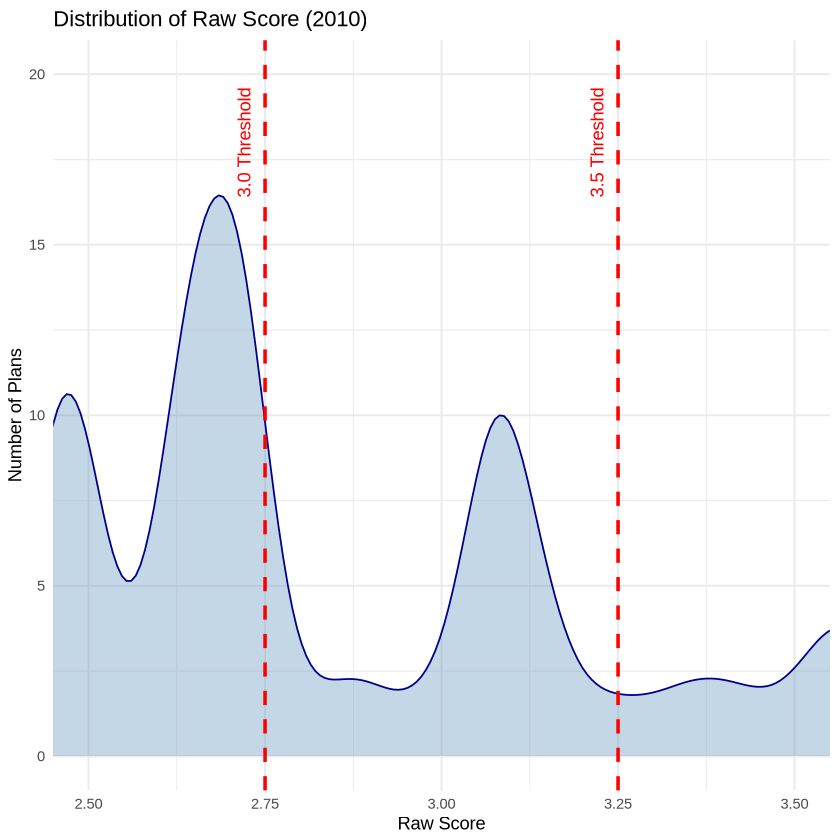
Note: *p<0.1; **p<0.05; ***p<0.01

Question 7



This bandwidth sensitivity analysis shows that RD estimates are robust to the choice of bandwidth. Regardless of window size (0.1 to 0.15), the estimated effect of a star rating increase remains statistically insignificant.

Question 8

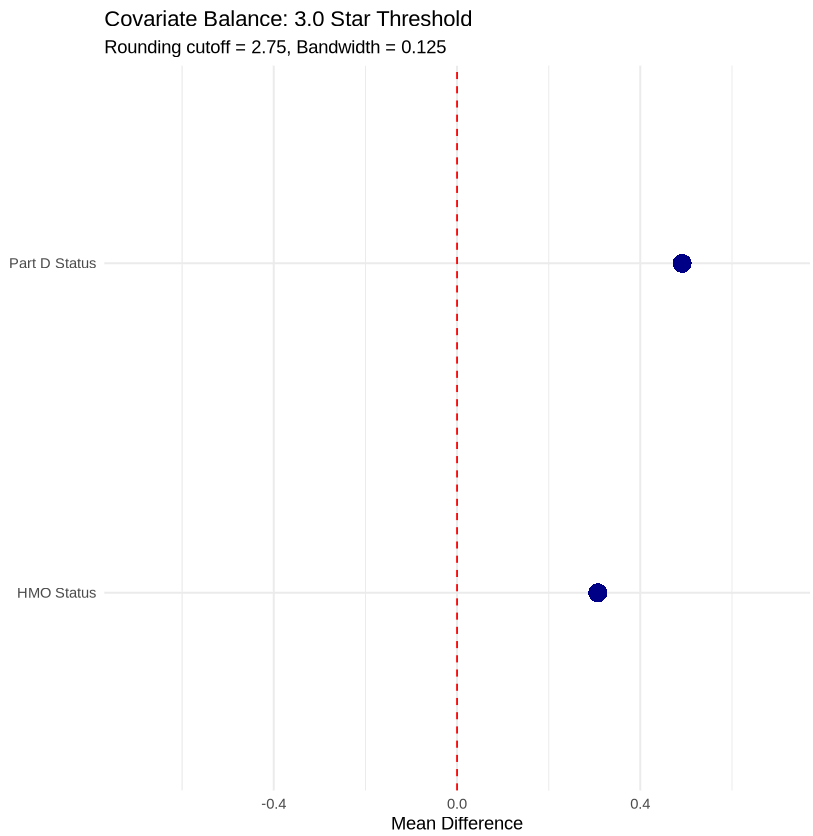


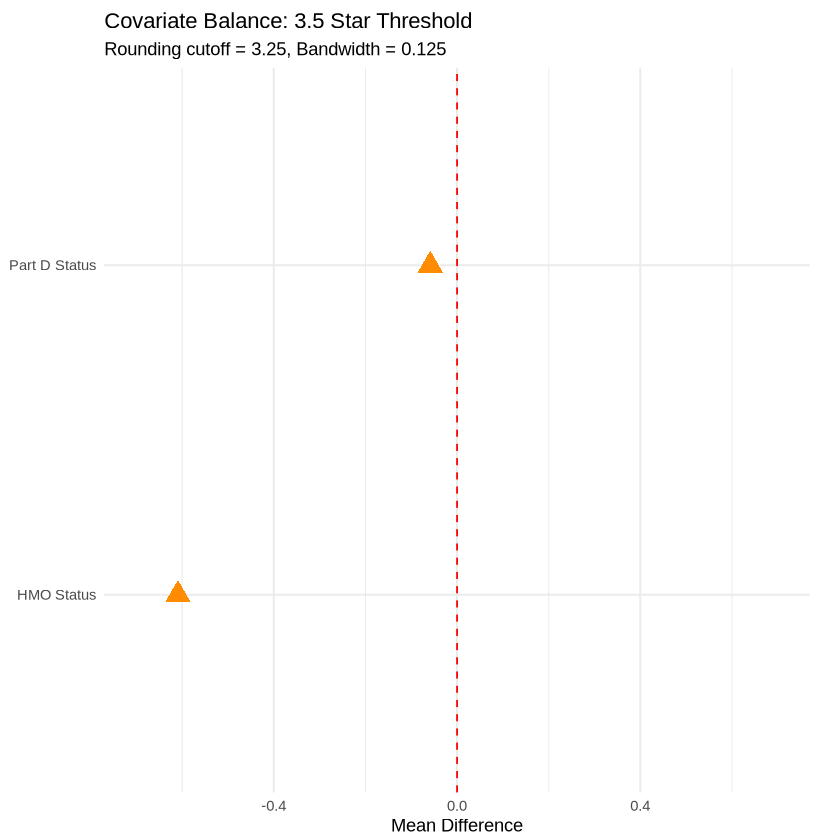
The density plot shows a continuous distribution across the 2.75 and 3.25 thresholds (at which scores are rounded up to 3 and 3.5, respectively). There is no bunching or a discontinuous jump in the density of plans immediately following the rounding cutoffs. This supports the earlier findings that contracts did not strategically manipulate quality metrics to cross the thresholds in 2010.

Question 9

```
Table 1: Covariate Balance (2010)
=====================================================================
                                Dependent variable:
                  ---------------------------------------------------
                    HMO (3.0)   Part D (3.0)    HMO (3.5)    Part D (3.5)
                       (1)          (2)           (3)           (4)
---------------------------------------------------------------------
Mean Difference      0.307        0.491         -0.609        -0.058
                  t = 19.481*** t = 15.598*** t = -13.797*** t = -1.751*

---------------------------------------------------------------------
Observations         5,984        5,984         2,172         2,172
=====================================================================
Note:                                       *p<0.1; **p<0.05; ***p<0.01
                                                  Bandwidth = 0.125
```


Covariate Balance: 3.0 Star Threshold
Rounding cutoff = 2.75, Bandwidth = 0.125
Part D Status
HMO Status
-0.4
0.0
0.4
Mean Difference


Covariate Balance: 3.5 Star Threshold
Rounding cutoff = 3.25, Bandwidth = 0.125
Part D Status
HMO Status
-0.4
0.0
0.4
Mean Difference

Question 10

Overall conclusion: Looking at results from questions 5-9, we can conclude that increasing star rating had no statistically significant effect on MA enrollments in 2010.

Questions 6/7: RD estimates at the 3.0 and 3.5-star thresholds were statistically insignificant across all tested bandwidths (0.1 to 0.15). This demonstrates that jumping to a higher star rating did not cause a discontinuous boost in enrollment.

Question 9: The T-statistic table reveals large imbalance. Plans just above 2.75 were 30.7% more likely to be HMOs ($t = 19.481$) and 49.1% more likely to offer Part D ($t = 15.598$). Both HMO status and Part D status show statistically significant discontinuities at the 2.75 and 3.25 thresholds. This indicates that plans above the rounding cutoffs were systematically different from those just below. Therefore, any observed effect on enrollment cannot be attributed only to the star rating, since these underlying plan characteristics likely confound the estimate.